THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE FULL SYLLABUS PAPER SET 08

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A claim says ‘food becomes harmful during an eclipse’. The best scientific response is to: 1

(a) ask what mechanism and measurable change could make it harmful
(b) accept it because it is old
(c) reject it without asking anything
(d) accept it because many people share it

2. Estimating the rice needed for a family of four for a month: the answer 100 g is unreasonable because: 1

(a) estimates must be exact **(b)** it is far too little for a whole month
(c) it is exactly right **(d)** rice cannot be weighed

3. Which pair is a mixture of immiscible liquids? 1

(a) alcohol and water **(b)** oil and water
(c) acetone and water **(d)** vinegar and water

4. 18 g of a solute is dissolved in water to make 200 mL of solution. Its mass by volume percentage (% m/v) is: 1

(a) 8.26 % **(b)** 4.5 % **(c)** 18 % **(d)** 9 %

5. The chemical formula of magnesium oxide is: 1

(a) MgO_2 **(b)** MgO **(c)** Mg_2O_2 **(d)** Mg_2O

6. The correct name of the covalent compound SF_6 is: 1

(a) sulfur hexafluoride **(b)** disulfur hexafluoride
(c) sulfur hexafluorine **(d)** sulfur fluoride

7. In the gold foil experiment most α-particles passed straight through. This showed that: 1

(a) most of the atom is empty space **(b)** α-particles are neutral
(c) gold is radioactive **(d)** atoms are solid spheres

8. A vehicle moving at 10 m s^{-1} is brought to rest with a uniform retardation of 2 m s^{-2}. The stopping distance is: 1

(a) 50 m **(b)** 5 m **(c)** 12.5 m **(d)** 25 m

9. A speed of 108 km h^{-1} in SI units is: 1

(a) 60 m s^{-1} **(b)** 30 m s^{-1} **(c)** 388.8 m s^{-1} **(d)** 36 m s^{-1}

10. A force of 60 N produces an acceleration of 3 m s^{-2} in a body. Its mass is: 1

(a) 20 kg **(b)** 3 kg **(c)** 180 kg **(d)** 40 kg

11. A gun recoils when a bullet is fired. This is explained by: 1

(a) law of gravitation **(b)** Newton's third law
(c) Hooke's law **(d)** Newton's first law

12. A machine does 2000 J of work in 8 s. Its power is: 1

(a) 500 W **(b)** 250 W **(c)** 16000 W **(d)** 0 W

13. The minimum time gap needed to hear an echo separately is: 1

(a) 1 s **(b)** 0.01 s **(c)** 0.1 s **(d)** 0.5 s

14. The functional segments of DNA are called: 1

(a) chromatin **(b)** genes **(c)** plastids **(d)** ribosomes

15. Growth of an organism is mainly due to: 1

(a) increase in cell size only **(b)** increase in the number of cells
(c) death of cells **(d)** decrease in cell size

16. Growth in the length of plant roots and shoots is due to: 1

(a) lateral meristem **(b)** epidermis
(c) apical meristem **(d)** intercalary meristem

17. Lichens are used as bioindicators of: 1

(a) rainfall **(b)** air quality
(c) water temperature **(d)** soil fertility

18. Reproduction by budding is seen in: 1

(a) potato and ginger **(b)** amoeba and bread mould
(c) yeast and hydra **(d)** fish and frog

19. The baby gets nutrition and oxygen from the mother's blood through the: 1

(a) cervix **(b)** ovary **(c)** oviduct **(d)** placenta

20. The layer of the atmosphere where weather occurs is the: 1

(a) stratosphere **(b)** mesosphere **(c)** exosphere **(d)** troposphere

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. Name two branches of science involved in studying a pressure cooker and one idea from each. 2

22. Calculate the molecular mass of CO_2 and NH_3 (C=12, O=16, N=14, H=1). 2

23. What does the area under a velocity–time graph represent? How is it found for constant velocity? 2

24. Why does a fielder draw his hands back while catching a fast ball? 2

25. Name the chief constituent of the plant cell wall. Give one use of it in our diet. 2

26. Differentiate between parenchyma and collenchyma. 2

27. Name the four whorls of a complete flower. 2

28. Define albedo. Give one high-albedo and one low-albedo surface. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. What are alloys? Why are they useful? Give two examples. 3

30. State three uses of isotopes. 3

31. A person raises a 30 cm step with a 40 cm wide ramp. Find the length of the ramp and the mechanical advantage. 3

32. A source produces a wave of wavelength 3.44 m at a speed of 344 m s^{-1}. Find the time period. 3

33. Describe F. C. Steward's experiment on carrot phloem cells. 3

34. Name the five classes of vertebrates with an example for each. 3

35. A student burns 10 g of ethanol in an open beaker and finds the mass decreased. Is the law of conservation of mass violated? Explain. 3

OR

Show the formation of a chlorine molecule by sharing of electrons. 3

36. Describe the electromagnetic spectrum from long to short wavelengths. 3

OR

Why is UV radiation harmful and useful? 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. (i) What is nuclear energy? Name one use. 5
(ii) Name the scientist called the Father of the Indian Nuclear Programme.
(iii) How does a nuclear power plant produce electricity?

OR

Case study: A student reacted zinc with dilute hydrochloric acid in an open test tube and found the mass decreased. 5
(i) Name the products.
(ii) Why did the mass decrease?
(iii) How can she modify the set-up to verify the conservation of mass?
(iv) What result should she get?

38. (i) What is friction? 5
(ii) Describe the experiment to compare friction on different surfaces.
(iii) How does friction help us to walk?

OR

(i) State Newton's second law in terms of momentum. 5
(ii) Define momentum.
(iii) A 2 kg ball moving at 5 m s^{-1} is stopped in 0.5 s; find the force.

39. (i) Explain programmed cell death. 5
(ii) How do cancer cells differ from normal cells?
(iii) Why is cell division regulated?

OR

Case study: A student wants to estimate the size of cells under the microscope: a field of 4 mm shows 20 cells in a row; eyepiece 10X and objective 10X. 5

(i) Find the size of one cell.

(ii) What is the total magnification?

(iii) How big does one cell appear?

(iv) Which part of the microscope gives clarity?

40. (i) Compare external and internal fertilisation. 5

(ii) Why is the survival rate higher in internal fertilisation?

(iii) What is IVF?

OR

(i) Why is responsible behaviour important in adolescence? 5

(ii) Name two STIs and prevention.

(iii) Describe three contraceptive methods.

END OF QUESTION PAPER **Total: 80 Marks**

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